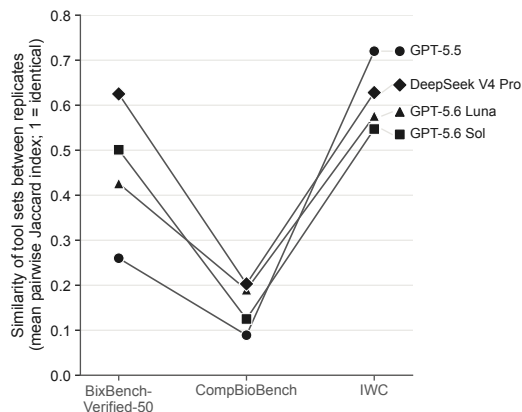


a Open-ended code runs spread across more command-line software than Galaxy runs

	Open-ended code	Galaxy	Open-ended code	Galaxy	Open-ended code	Galaxy
	BixBench-Verified-50		CompBioBench		IWC	
phykit	12	14	-	-	-	-
DESeq2 or PyDESeq2	17	7	-	-	20	11
gseapy	8	6	-	1	-	-
samtools	4	2	28	10	40	12
bcftools	2	2	12	5	8	-
bedtools	-	-	14	11	13	2
anndata	2	-	10	7	10	7
scanpy	1	-	9	4	10	-
datamash	-	8	-	9	1	-
bwa	2	2	14	6	26	4
fastp	-	-	5	-	28	20
bowtie2	-	-	14	3	22	20
minimap2	1	-	19	2	27	1
MACS2 or MACS3	-	-	5	1	10	8
edgeR	4	-	-	-	14	10
dada2	-	-	-	-	10	9

Cell value: % of runs using the software (three GPT configurations shared by all benchmarks).
Open-ended code: software named in shell commands. Galaxy: installed tools in job records.

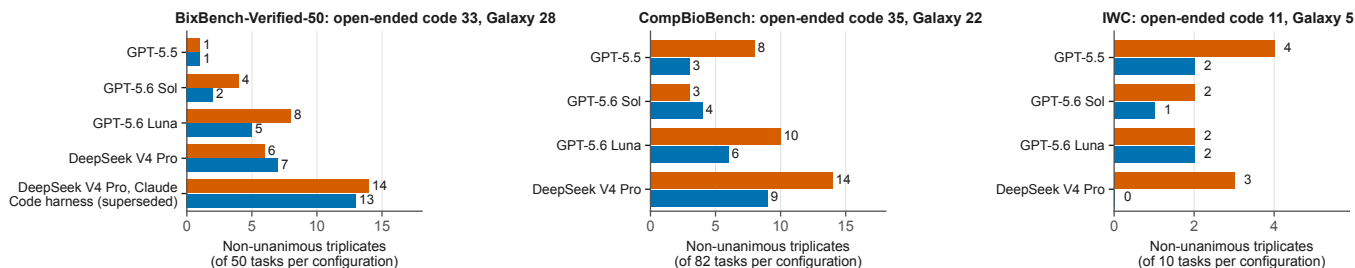
b Replicate tool sets in Galaxy agree most on workflow-derived tasks



c Galaxy produced fewer non-unanimous triplicates than open-ended code in every benchmark

Open-ended code Galaxy

A triplicate is one task run three times with one configuration in one environment; it is non-unanimous when 1 or 2 of the 3 replicates succeeded (BixBench-Verified-50: accepted by the evaluator; CompBioBench: matched the answer given by at least 20 of 25 runs, 82 tasks; IWC: replicate scores differed by more than 0.05).



d Replicates diverged for different reasons in each environment (BixBench-Verified-50)

Hand-written methods and software-version differences: 67% of open-ended-code divergences, 14% of Galaxy divergences. Galaxy interface traps: 17% of Galaxy divergences, none in open-ended code.

